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FREE GAS FOR TESTING THE DApp (SEPOLIA TEST ETH)

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This app runs on the Ethereum SEPOLIA TESTNET. "Gas" is the small fee paid to the network for each blockchain transaction (uploading a file, sharing it, revoking access...). You pay it with SEPOLIA test ETH, which is FREE – it has no real-money value. This guide shows visitors how to get enough of it to test the app.

Quick summary: get MetaMask -> switch to Sepolia -> claim free test ETH from a faucet -> you are ready to test.

WHICH ACTIONS NEED GAS?

Costs a little test ETH (0.0001-0.0002 ETH per call):

- Upload a file (register it on-chain)
- Grant / revoke access to someone
- Any transaction that writes to the contract

Free (no ETH needed):

- Connecting your wallet
- Viewing "My Files", "Shared With Me", "Activity Log"
- Downloading files you have access to

So a few 0.1 ETH is enough for many dozens of test transactions.

STEP 1 - INSTALL META MASK

1. Install the MetaMask browser extension (Chrome/Firefox/Edge/Brave): <https://metamask.io/download/>
2. Open it and create a new wallet (or log into an existing one).
3. IMPORTANT: it is a testnet, so do NOT fund it with real money. Never use an account that holds real (mainnet) funds for this.

STEP 2 - ADD THE SEPOLIA NETWORK TO META MASK

When you open the app and click "Connect Wallet", the DApp asks MetaMask to add/switch to Sepolia automatically - just approve it.

To do it manually instead:

1. Open the MetaMask popup.
2. Open the network list / settings (the new MetaMask version moved the network menu - look for the network chip near your token list).
3. Find "Sepolia" and select it. If it is not listed, use "Add network":

Network name:	Sepolia
New RPC URL:	https://rpc.sepolia.org
Chain ID:	11155111
Currency symbol:	SEPOLIAETH
Block explorer:	https://sepolia.etherscan.io
4. Save. MetaMask now shows your (zero) Sepolia balance.

STEP 3 - GET FREE SEPOLIA TEST ETH FROM A FAUCET

A "faucet" is a website that sends free test ETH to any wallet address. You need your wallet address:

- In MetaMask, click your account name -> "Copy address".
- It looks like: 0x123f9b48a65FA8836e620D1573AE63815C6D7D01

Pick any faucet below, paste that address, and submit:

1. Alchemy Sepolia faucet (needs a free Alchemy login)
<https://www.alchemy.com/faucets/ethereum-sepolia>
-> ~0.5 SEPOLIAETH, 1 claim every 24h
2. Google Cloud Web3 faucet (free, quick, no crypto login)
<https://cloud.google.com/application/web3/faucet/ethereum/sepolia>
-> ~0.5 SEPOLIAETH, 1 claim every 24h
3. QuickNode Sepolia faucet (free)
<https://faucet.quicknode.com/ethereum/sepolia>
-> small amount, 1 claim per day
4. Chainlink Sepolia faucet (gives ETH + LINK, needs GitHub login)
<https://faucets.chain.link/sepolia>
5. Community faucets (may ask for a social login)
<https://www.sepoliafaucet.io/>

Note: faucets send to the ADDRESS you paste, so double-check you copied the correct one. If you paste the wrong address, the ETH goes to that address and cannot be recovered.

STEP 4 - CHECK YOUR BALANCE

1. Make sure MetaMask is still on the SEPOLIA network (Step 2).
2. Wait 30 seconds to a few minutes for the faucet transaction.
3. Open the MetaMask popup - your balance should now show test ETH (e.g. 0.5 SEPOLIAETH).
4. If it still shows 0, you are probably viewing the wrong network in MetaMask (the default view shows Ethereum Mainnet, which is empty). Switch to Sepolia and check again.

STEP 5 - USE THE APP

1. Open the app URL and click "Connect Wallet".
2. Approve the network switch to Sepolia if MetaMask asks.
3. Upload a file, download it, share it with another address, etc.
4. MetaMask will show a transaction confirmation for each on-chain write - review and confirm it.

When a transaction is pending, it usually confirms within 30 seconds. Sepolia is a public testnet, so occasionally the network is busy and a transaction takes a minute - that is normal.

RUNNING LOW? REFILLING

- You can claim from a DIFFERENT faucet if the first one has a daily limit.
- After 24h you can claim from the same faucet again.
- If you created a brand-new wallet for testing, the old faucet claims still went to the old address - claim fresh ETH to the new address.

IMPORTANT NOTES

- SEPOLIA test ETH has NO real value. Never pay real money for it.
- Never paste a private key or secret recovery phrase into a faucet or anywhere online - faucets only need your public wallet ADDRESS.
- The same Sepolia RPC setup for developers is explained in `RPC_ENDPOINT_SETUP_GUIDE.txt`.

- This project only runs on Sepolia; there is no mainnet deployment, so no real ETH is ever required.

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